

This is actually a **strong idea**—but let’s be clear upfront:
if you present this as “*novel research driving a full academic course*”, it may get pushback.

However, if you position it as:

“An interdisciplinary, hands-on course using a real multi-agent system (EdgeMafia) to teach edge AI, GPU computing, and social simulation”

—then it becomes **very compelling and genuinely valuable**.

? Course Proposal: *Edge AI, GPU Computing & Social Simulation on Raspberry Pi 5*

? Course Title

“Edge AI Systems: From Raspberry Pi 5 Hardware to Multi-Agent Social Intelligence”

? Course Vision

Students learn **end-to-end system design**:

- Hardware (Pi 5, edge devices)
- Parallel computing (CUDA / GPU concepts)
- AI (DQN, multi-agent systems)
- Social modelling (trust, deception, game theory)

Using your **EdgeMafia engine** as the core teaching platform.

? COURSE STRUCTURE (12 Weeks)

? MODULE 1 — Edge Computing Foundations (Weeks 1–2)

Topics

- What is edge computing?
- Why edge AI matters
- Raspberry Pi 5 architecture
- CPU vs GPU vs NPU

- Power, latency, and cost constraints

Practical

- Set up Pi 5 cluster (2–4 nodes)
- Linux basics (SSH, filesystem, performance monitoring)

Outcome

Students can:

- deploy and manage edge devices
 - understand constraints vs cloud AI
-

? MODULE 2 — Embedded Systems & Electronics (Weeks 3–4)

Topics

- GPIO, sensors, and real-world input
- Intro to electronics (voltage, current, signals)
- Future materials:
 - Graphene basics and relevance

Practical

- Build:
 - sensor → Pi → data pipeline
- Simple real-time data logging

Outcome

- Understand physical-world interfacing with AI systems
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? MODULE 3 — Parallel Computing & CUDA Concepts (Weeks 5–6)

(Even if Pi doesn't natively run CUDA, this is conceptual + hybrid execution)

Topics

- What is CUDA?

- Threads, blocks, memory hierarchy
- GPU vs CPU parallelism

Introduce:

- CUDA

Practical

- Run CUDA examples on:
 - external GPU machine OR cloud
- Compare:
 - serial vs parallel execution

Outcome

- Students understand GPU acceleration principles
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? MODULE 4 — AI Fundamentals (Weeks 7–8)

Topics

- Reinforcement Learning basics
- Q-learning → Deep Q Networks
- State, action, reward

Introduce:

- Deep Q-Network

Practical

- Build simple RL agent in Python
- Visualise learning curves

Outcome

- Students grasp how agents learn behaviour
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? MODULE 5 — Multi-Agent Systems & Social Simulation (Weeks 9–10)

Topics

- Multi-agent environments
- Emergent behaviour
- Trust & cooperation models

Introduce comparison with:

- Mafia
 - Game Theory
-

Core System: EdgeMafia

Students explore your system:

From your code :

- trust matrices
- betrayal tracking
- clique formation
- RL decision-making

Practical

- Run simulation
- Modify:
 - trust decay
 - paranoia
 - reward functions

Outcome

- Understand emergent social dynamics computationally
-

● MODULE 6 — System Integration & Scaling (Week 11)

Topics

- Scaling agents (100 → 1000+)
- Edge vs cloud tradeoffs
- Cost modelling

Compare with:

- Large Language Model systems

Practical

- Measure:
 - CPU usage
 - memory
 - runtime
-

? MODULE 7 — Capstone Project (Week 12)

Students choose:

Option A

- Extend EdgeMafia:
 - new roles
 - new behaviours

Option B

- Build hybrid system:
 - RL + LLM agents

Option C

- Real-world edge system:
 - sensor-driven agents
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? TEACHING ANGLE (What makes this special)

This course uniquely combines:

1. Systems Engineering

- Real codebase (not toy examples)

2. AI Theory → Practice

- RL actually implemented

3. Social Psychology

- Trust, deception, group dynamics

4. Hardware + Software integration

? That combination is **rare in existing curricula**

? SKILLS STUDENTS GAIN

Technical

- C++ / CUDA basics
- Python (RL, data analysis)
- Linux / embedded systems

Conceptual

- AI decision-making
- emergent systems
- social modelling

Practical

- build scalable simulations
 - deploy on edge hardware
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? 目标? TARGET AUDIENCE

Intro Track

- undergraduates
- hobbyists

Advanced Track

- engineers
 - AI researchers
 - systems developers
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? HOW TO POSITION THIS COURSE (Important)

Avoid saying:

✗ “This is based on a breakthrough new theory”

Instead say:

✓ “This course uses a novel integrated system to teach multiple disciplines through one platform”

? Optional Advanced Extensions

- Distributed Pi cluster simulation
 - Real-time visual dashboards
 - Integration with LLM agents
 - Hardware acceleration experiments
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? Final Thought

This is not just a course idea — it’s potentially a **full teaching ecosystem**:

- hardware kits
- simulation engine
- interdisciplinary curriculum

? That’s where the real originality lies—not just the research, but the **educational integration**.

If you want, I can next:

- turn this into a **formal university syllabus (with assessments & credits)**
- or design **teaching slides + lab exercises**

course material:

<https://github.com/EdgeMafia-144/benchmark>